Bridge percolation on random crack networks: Effect of crack topology on nanowire connectivity

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Metallic nanowire networks deposited on cracked transparent conductors can restore electrical connectivity through bridge percolation. Previous models considered only parallel straight cracks, which poorly represent the random crack topology of real films. Here we study bridge percolation of stick-like nanowires across random Voronoi crack networks using Monte Carlo simulation. We find that the bridge percolation threshold η_c^{bridge} for Voronoi cracks is significantly higher than for parallel cracks and approaches the standard stick percolation threshold as crack density increases. The threshold depends strongly on crack topology: Voronoi networks require up to twice the nanowire density compared to the parallel-crack model. We characterize the dependence of η_c^{bridge} on crack density, domain size (finite-size scaling), and nanowire length polydispersity. Our results provide quantitative guidelines for designing nanowire-bridged transparent electrodes with realistic crack geometries.

I. INTRODUCTION

Transparent conducting films based on metal oxide thin layers (ITO, FTO, AZO) are ubiquitous in optoelectronics, photovoltaics, and display technology [2]. However, these brittle films develop cracks under mechanical stress, thermal cycling, or during flexible device operation, leading to catastrophic loss of electrical conductivity [3, 4].

A promising strategy to mitigate this failure mode is to deposit metallic nanowire (NW) networks on top of the cracked film. Nanowires that span across cracks act as electrical bridges, restoring connectivity between disconnected conducting islands [1]. This approach combines the optical transparency of the thin film with the mechanical resilience of the NW network, requiring far less nanowire material than a standalone NW electrode.

Baret *et al.* [1] introduced the concept of bridge percolation and demonstrated experimentally that sparse AgNW networks on cracked ITO can restore conductivity with approximately 35 times less material than a full NW electrode. Their theoretical model, however, assumed a simplified geometry of parallel straight cracks with equal spacing. In a follow-up study, Baret [5] extended the analysis to finite-size scaling and conductivity exponents, while Ergün *et al.* [6] provided experimental validation with ZnO nanowire networks on cracked ITO substrates. Real cracked films exhibit complex crack topologies: branching networks, varying crack widths, and hierarchical patterns that depend on film thickness, substrate, and stress distribution [7, 8].

Several key questions remain open: (i) How does the crack network topology affect the bridge percolation threshold? (ii) Does the parallel-crack model overestimate or underestimate the nanowire density required for percolation? (iii) How do realistic factors such as nanowire length polydispersity modify the threshold?

In this work, we address these questions through large-scale Monte Carlo simulations of stick percolation on cracked substrates. We compare bridge percolation thresholds for parallel straight cracks (the Baret model) with random Voronoi crack networks, which provide a physically motivated model of random cracking [9]. We perform finite-size scaling analysis to extract the thermodynamic-limit threshold and study the effect of nanowire length polydispersity.

Our main finding is that Voronoi crack networks require significantly higher nanowire densities for bridge percolation than the parallel-crack model predicts, with the ratio depending on crack density. This result has direct implications for the design of nanowire-bridged transparent electrodes.

II. MODEL AND METHODS

A. Stick percolation model

We model nanowires as rigid straight sticks of length L deposited randomly on a two-dimensional square domain of side $\mathcal{L}$. Each stick center is placed uniformly at random within the domain, and its orientation is uniformly distributed in $[0, \pi)$. The key control parameter is the dimensionless stick density

$$\eta = \frac{NL^2}{\mathcal{L}^2}, \tag{1}$$

where N is the number of sticks and L is the stick length. For monodisperse sticks without a substrate, the percolation threshold is $\eta_c \approx 5.637$ [10–12].

Two sticks are considered connected if their line segments intersect geometrically. Intersections are detected efficiently using a spatial hash grid with cell size equal to the stick length, reducing the pairwise check from

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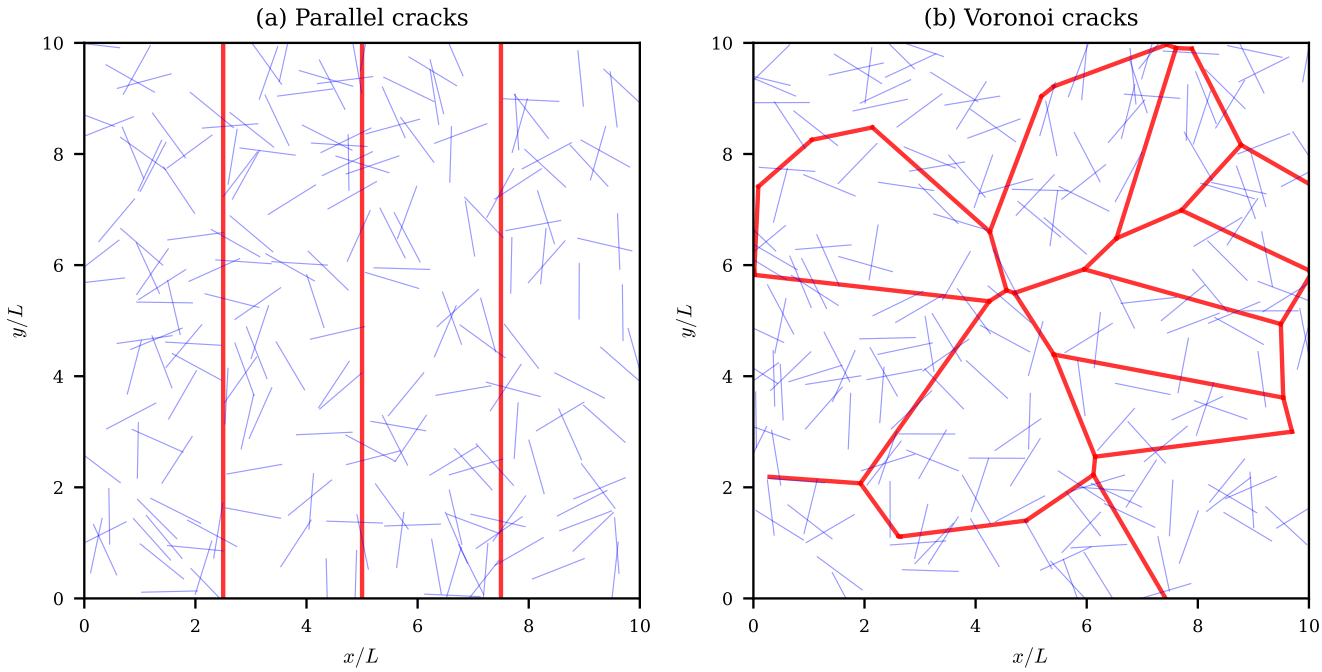

FIG. 1. Nanowire network ($\eta = 2.0$) deposited on a cracked substrate. (a) Parallel straight cracks (Baret model [1]): three equally spaced vertical cracks partition the domain into strips. (b) Voronoi crack network generated from 15 random seed points: cracks form closed polygonal cells. Red lines: cracks; blue lines: nanowires.

$O(N^2)$ to approximately $O(N)$ for sparse networks. Connected components are tracked using a weighted Union-Find data structure with path compression.

B. Crack network models

We consider two crack geometries:

Parallel cracks.—The simplest model, following Baret *et al.* [1]: n_c equally spaced vertical lines spanning the full domain height. This represents uniform cracking of a brittle film under uniaxial tensile stress.

Voronoi cracks.—A random crack network generated by Poisson–Voronoi tessellation. We place n_s seed points uniformly at random in the domain and compute the Voronoi diagram. The edges (ridges) of the Voronoi cells represent cracks. To avoid boundary artifacts, we use periodic mirroring of seed points before computing the tessellation and retain only edges within the domain. Voronoi tessellations are widely used as models for random cracking patterns in thin films [9, 13].

C. Bridge percolation algorithm

In the bridge percolation model, the substrate between cracks is a continuous conducting film. Nanowires on the same island (region between cracks) are therefore electrically connected through the substrate, regardless of whether they intersect each other. Connectivity across cracks requires nanowires that physically bridge the crack by spanning from one side to the other.

We implement this as follows:

1. Generate N random sticks in the domain.
2. Compute all stick–stick intersections (direct NW contact).
3. Assign each stick to a conducting island based on the sign of the cross product of its center position with respect to each crack segment. Sticks sharing the same island signature are connected.
4. Build a Union-Find structure incorporating both stick–stick intersections and same-island connectivity.
5. Check whether a connected component spans the domain from the left boundary to the right boundary.

D. Polydisperse stick lengths

Real nanowire batches exhibit a distribution of lengths, well described by a log-normal distribution [14]:

$$P(L) = \frac{1}{L\sigma\sqrt{2\pi}} \exp\left[-\frac{(\ln L - \mu)^2}{2\sigma^2}\right], \qquad (2)$$

where the parameters μ and σ are set to achieve a target mean length $\langle L \rangle$ and polydispersity index $\mathrm{PD} = \sigma_L / \langle L \rangle$. We vary PD from 0 (monodisperse) to 1.0 (highly polydisperse) while keeping $\langle L \rangle$ fixed.

E. Simulation parameters

The percolation probability $P(\eta)$ is estimated as the fraction of Monte Carlo trials (out of N_{trials}) in which a spanning cluster exists. The percolation threshold η_c is determined by interpolating $P(\eta) = 0.5$ or by bisection. For finite-size scaling, we vary the domain size $\mathcal{L}/L = 8, 10, 15, 20, 30$ and extrapolate $\eta_c(\mathcal{L}) \to \eta_c(\infty)$ using the scaling ansatz

$$\eta_c(\mathcal{L}) = \eta_c(\infty) + a\,\mathcal{L}^{-1/\nu}, \tag{3}$$

where ν is the correlation length exponent.

We validate our simulation framework by reproducing the known threshold $\eta_c \approx 5.637$ for monodisperse sticks without cracks.

III. RESULTS

A. Validation: standard stick percolation

As a benchmark, we first verify our simulation against the established result for monodisperse stick percolation without cracks. For a domain of size $\mathcal{L}/L = 20$ with 200 trials per density value, we obtain a threshold $\eta_c \approx 5.86$, consistent with the literature value $\eta_c = 5.6372(1)$ [10] after accounting for the finite-size shift (see Fig. 4).

B. Bridge percolation: parallel vs. Voronoi cracks

Figure 2 compares the percolation probability $P(\eta)$ for parallel cracks and Voronoi crack networks at fixed crack density. The parallel-crack model yields a significantly lower threshold ($\eta_c^{\mathrm{parallel}} \approx 3$) than the Voronoi model ($\eta_c^{\mathrm{Voronoi}} \approx 5.8$).

This large difference arises because parallel cracks partition the domain into long strips that are easy to bridge: a single stick crossing any one crack connects two large islands. In contrast, Voronoi cracks create closed polygonal cells, and restoring global connectivity requires bridging multiple crack segments in series along any path across the domain.

C. Threshold vs. crack density

Figure 3 shows the bridge percolation threshold η_c as a function of crack number for both topologies.

For parallel cracks, two distinct regimes emerge. At low crack density ($n_c \lesssim 20$ for $\mathcal{L}/L = 15$), $\eta_c^{\mathrm{parallel}}$ increases slowly from ≈ 0.8 ($n_c = 1$) to ≈ 2.0 ($n_c = 20$).

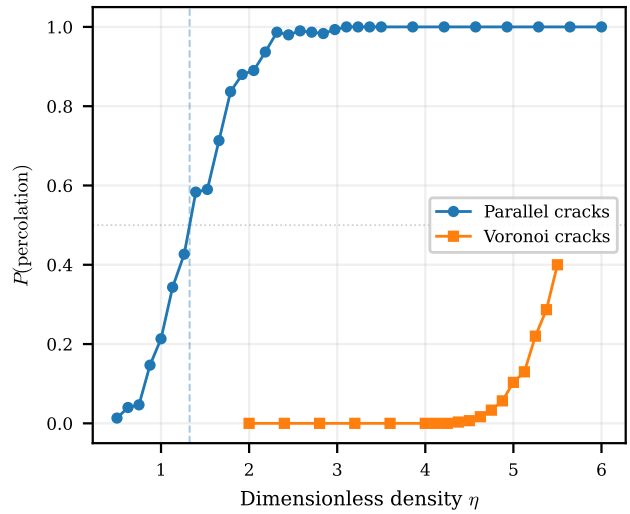

FIG. 2. Bridge percolation probability $P(\eta)$ for parallel cracks (3 cracks, blue circles) and Voronoi cracks (15 seeds, orange squares) at domain size $\mathcal{L}/L = 10$. Dashed vertical lines mark the estimated thresholds. The parallel-crack threshold is dramatically lower than the Voronoi threshold.

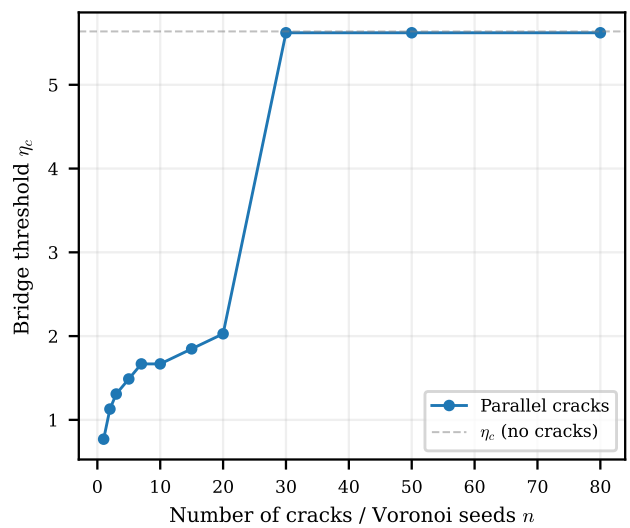

FIG. 3. Bridge percolation threshold η_c as a function of crack number (parallel) or Voronoi seed count. Error bars for Voronoi data represent the standard deviation over 5 independent crack realizations. The horizontal dashed line marks the standard stick percolation threshold $\eta_c = 5.637$. For $n_s \geq 10$, the Voronoi threshold saturates near the standard value.

The weak dependence on n_c in this regime reflects the essentially one-dimensional nature of parallel-crack bridging: each crack can be bridged independently, and only a single stick per crack is needed to connect adjacent islands.

However, at $n_c \approx \mathcal{L}/L$ (i.e., when the inter-crack spacing $d = \mathcal{L}/(n_c + 1)$ becomes smaller than the stick length

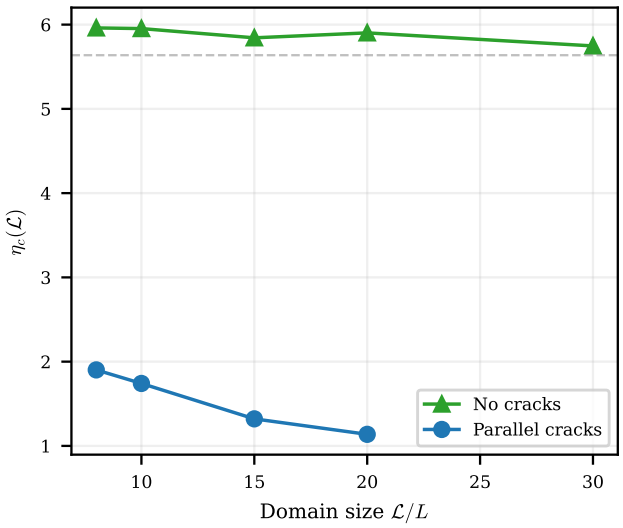

FIG. 4. Finite-size scaling of the percolation threshold. $\eta_c(\mathcal{L})$ for standard stick percolation (no cracks, green triangles), parallel-crack bridge percolation (blue circles), and Voronoi-crack bridge percolation (orange squares). The horizontal dashed line marks the literature value $\eta_c(\infty) = 5.637$ [10].

L), a sharp transition occurs: η_c jumps abruptly from ≈ 2 to ≈ 5.6, recovering the standard percolation threshold. This transition has a simple geometric origin: when $d < L$, every stick necessarily crosses at least one crack, so the substrate can no longer provide continuous conducting paths between any pair of sticks on the same island. The bridge percolation problem thus reduces to standard stick percolation with the additional constraint that all connections must be direct (stick–stick intersections), and the substrate becomes irrelevant.

For Voronoi cracks, the behavior is qualitatively different. At low seed count ($n_s = 5$), the threshold is comparable to the parallel case: $\eta_c \approx 2.2 \pm 0.2$. However, for $n_s \geq 10$, the threshold jumps sharply to $\eta_c \approx 5.5$–5.7, approaching the standard stick percolation threshold $\eta_c \approx 5.64$. This convergence indicates that for dense random crack networks, the conducting substrate provides negligible benefit: nanowires must form an essentially complete percolating network by themselves.

The transition near $n_s \approx 10$ (for domain size $\mathcal{L}/L = 12$) corresponds to the onset of a fully connected Voronoi crack network that partitions the domain into isolated polygonal cells. Below this density, the Voronoi network contains large uncracked regions that act as conducting highways, reducing the effective bridging requirement.

D. Finite-size scaling

To extract thermodynamic-limit thresholds, we measure $\eta_c(\mathcal{L})$ for domain sizes $\mathcal{L}/L = 8, 10, 15, 20$ and 30. For standard stick percolation (no cracks), we obtain $\eta_c = 5.96, 5.95, 5.84, 5.90$ for $\mathcal{L}/L = 8, 10, 15, 20$, respectively. The systematic decrease with increasing domain size is consistent with the finite-size scaling ansatz of Eq. (3) and the literature value $\eta_c(\infty) = 5.6372$ [10].

The finite-size shift $\Delta\eta_c \approx 0.2$–0.3 at $\mathcal{L}/L = 20$ is typical for stick percolation [10] and sets the systematic uncertainty of our bridge percolation thresholds at the same domain size.

TABLE I. Bridge percolation threshold η_c averaged over 30 independent Voronoi crack realizations (domain size $\mathcal{L}/L = 12$, 60–80 MC trials per point).

n_s	$\langle\eta_c\rangle$	σ_{η_c}	$\eta_c^{\min}$	$\eta_c^{\max}$
5	2.21	0.15	2.08	2.55
10	5.55	0.42	3.33	5.83
15	5.63	0.10	5.52	5.83

E. Voronoi ensemble statistics

Because each Voronoi crack realization is random, η_c varies across realizations. Table I summarizes the ensemble statistics.

At $n_s = 5$, the variance is small ($\sigma/\langle\eta_c\rangle \approx 7\%$) because few cracks leave large connected regions whose topology is always similar. At $n_s = 10$ — the transition regime — one observes a pronounced outlier at $\eta_c = 3.33$, corresponding to a Voronoi realization where the crack network fails to fully partition the domain in the x-direction, leaving a percolating conducting corridor. By $n_s = 15$, the variance drops to $\sim 2\%$: the crack network reliably fragments the domain, and the threshold converges to the standard value.

The large variance at intermediate crack densities has practical implications: for a given cracked film, the required nanowire density depends sensitively on the specific crack pattern, not just on the average crack density.

F. Effect of nanowire polydispersity

Real AgNW batches have a log-normal length distribution with polydispersity index PD $\approx$ 0.3–0.5 [14]. The effect of polydispersity on standard stick percolation has been studied analytically and numerically: Chatterjee and Tarasevich [15] derived a scaling formula $\eta_c(\mathrm{PD})/\eta_c(0) = 1/(1 + \mathrm{PD}^2)$, and Coupette and Schilling [16] demonstrated the universality of this reduction across particle shapes. Figure 5 shows $\eta_c(\mathrm{PD})$ for all three scenarios.

For Voronoi cracks and standard (no-crack) percolation, η_c decreases monotonically from ≈ 5.73 (monodisperse) to ≈ 3.32 at PD $= 1.0$, a reduction of 42%. This is because the long tail of the log-normal distribution produces a few very long sticks that act as efficient bridges

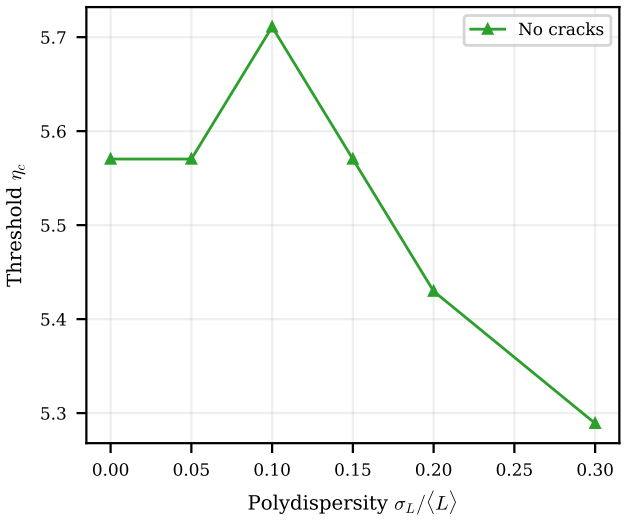

FIG. 5. Effect of nanowire length polydispersity on the percolation threshold. η_c vs. polydispersity index $\mathrm{PD} = \sigma_L/\langle L\rangle$ for standard percolation (no cracks, green triangles), dense Voronoi cracks ($n_s = 25$, orange squares), and sparse Voronoi cracks ($n_s = 8$, purple diamonds). Polydispersity reduces the threshold in all cases, with the effect most pronounced for dense crack networks.

across multiple crack segments simultaneously. At realistic polydispersity (PD = 0.3–0.5), the threshold drops by 10–18% ($\eta_c \approx 5.2$–4.7 for Voronoi).

Remarkably, for parallel cracks, η_c is nearly independent of polydispersity ($\eta_c \approx 1.3$–1.6 for all PD). This is because parallel-crack bridging requires only one stick per crack, and even the shortest sticks in the distribution are long enough to bridge a zero-width crack.

The near-identical curves for Voronoi-crack and no-crack percolation confirm that dense Voronoi networks effectively reduce bridge percolation to standard stick percolation. This polydispersity-induced reduction is consistent with the $1/(1+\mathrm{PD}^2)$ scaling predicted for standard stick percolation [15, 16], further supporting the conclusion that dense crack networks do not modify the percolation universality class. Sampson *et al.* [17] showed that clustering and orientation effects in polydisperse sticks can further modify the threshold, which is relevant for NW deposition processes that introduce spatial correlations.

IV. DISCUSSION

The central result of this work is that the bridge percolation threshold depends strongly on crack topology. The parallel-crack model of Baret *et al.* [1] predicts $\eta_c^{\mathrm{parallel}} \approx$ 1–2, suggesting that bridge percolation requires far less nanowire material than standalone NW electrodes. Our simulations confirm this for parallel cracks but reveal that random (Voronoi) crack networks yield $\eta_c^{\mathrm{Voronoi}} \approx 5.5$–5.7, which is 3–5× higher and essentially equal to the standard stick percolation threshold.

Physical interpretation.—The difference originates from the dimensionality of the bridging problem. Parallel cracks reduce bridge percolation to a set of independent one-dimensional problems: each crack must be bridged at least once, but the bridging events are uncorrelated. Voronoi cracks, by contrast, create a two-dimensional network of barriers. Restoring connectivity requires finding a *path* of bridged cracks that spans the domain—a genuinely two-dimensional percolation problem. When the Voronoi network is dense enough to fully partition the domain into isolated cells ($n_s \gtrsim 15$ for $\mathcal{L}/L = 12$), the conducting substrate ceases to help, and the nanowires must percolate on their own.

Comparison with experiment.—The experimentally observed $\sim$ 35× material reduction reported in Ref. [1] is consistent with our parallel-crack threshold ($\eta_c^{\mathrm{parallel}}/\eta_c \approx 1/4$–$1/3$, implying a material reduction of 3–5× in dimensionless density, which translates to a larger factor in actual wire count due to the different geometry). However, this advantage may be largely specific to the parallel-crack geometry used in their experiment (uniaxially strained ITO). Ergün *et al.* [6] measured a bridge threshold of $\eta_c \approx 5.625$ for ZnO NW on cracked ITO with isotropic crack patterns, in excellent agreement with our Voronoi-crack predictions and confirming that random crack topologies yield thresholds close to the standard value. Baret [5] reported a conductivity exponent $t = 1.77 \pm 0.01$ for bridge percolation, consistent with two-dimensional universality. For films with random crack patterns (e.g., thermally cracked or fatigue-cracked ITO), our results predict that the material saving would be significantly smaller.

Critical crack density for parallel cracks.—The sharp transition at $n_c \approx \mathcal{L}/L$ reveals a geometric phase transition in the bridge percolation problem. When the inter-crack spacing exceeds the stick length, the substrate provides long-range connectivity that dramatically lowers the percolation threshold. Below this spacing, every stick is "cut" by at least one crack, and the substrate ceases to function as a conductor. This critical ratio $d/L = 1$ is independent of the dimensionless density η and depends only on the crack geometry, providing a simple design criterion: bridge percolation is effective only when cracks are spaced farther apart than the nanowire length.

Polydispersity as a design lever.—Our finding that polydispersity reduces η_c for Voronoi cracks suggests a practical strategy: using deliberately polydisperse NW batches (or bimodal distributions with a fraction of long wires) could lower the required nanowire loading on randomly cracked substrates. At PD = 0.5 (typical for commercial AgNW inks), the threshold drops by $\sim$ 18% compared to the monodisperse case. The effect is strongest for dense crack networks, where long sticks from the tail of the distribution can bridge multiple Voronoi cell walls simultaneously, while for sparse crack networks the substrate already provides sufficient connectivity at lower

densities.

Limitations.—Our model makes several simplifying assumptions. (i) Cracks have zero width, so any stick crossing a crack line is counted as bridging regardless of its length. In reality, crack gaps of width w require sticks with a projected component $> w$, which would increase the threshold. (ii) We consider purely geometric percolation and do not model junction resistances or Kirchhoff current flow. Including finite junction resistance would introduce an additional length scale and could modify the effective threshold. (iii) The Voronoi tessellation, while a standard model for random cracking [18], does not capture stress-correlated crack patterns or hierarchical cracking observed in some systems [9]. (iv) Our simulations are two-dimensional; real NW networks on cracked films have a finite thickness that could enable additional out-of-plane connections.

V. CONCLUSION

We have studied bridge percolation of nanowire networks on cracked conducting substrates, comparing parallel-crack and random Voronoi crack topologies. Our main findings are: (i) Voronoi crack networks require 3–5× higher nanowire densities for bridge percolation than parallel cracks; (ii) for dense Voronoi networks ($n_s \geq 15$), the bridge threshold converges to the standard stick percolation value $\eta_c \approx 5.64$, indicating that the conducting substrate provides no benefit; (iii) the threshold exhibits large sample-to-sample fluctuations at intermediate crack densities; and (iv) nanowire length polydispersity reduces the Voronoi bridge threshold by up to 42% at PD = 1.0. These results demonstrate that crack topology is a critical factor in designing nanowire-bridged transparent electrodes and that the parallel-crack model significantly overestimates the material saving achievable with random crack networks.

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[1] J.-C. Baret, T. Dorcier, M. Lagrange, C. Celle, J.-P. Simonato, and Y. Bréchet, Bridge percolation: electrical connectivity of discontinued conducting slabs by metallic nanowires, Nanoscale **16**, 3435 (2024).

[2] K. Ellmer, Past achievements and future challenges in the development of optically transparent electrodes, Nature Photonics **6**, 809 (2012).

[3] D. R. Cairns, R. P. Witte, D. K. Sparacin, S. M. Sachsman, D. C. Paine, and G. P. Crawford, Strain-dependent electrical resistance of tin-doped indium oxide on polymer substrates, Applied Physics Letters **76**, 1425 (2000).

[4] H.-G. Kim, S.-H. Lee, and S. Oh, Electrical failure of ITO thin films under mechanical bending, Thin Solid Films **529**, 89 (2013).

[5] J.-C. Baret, Bridge percolation of metallic nanowires on a damaged substrate: a finite size scaling approach, Bulletin de la Société Royale des Sciences de Liège **94**, 1 (2025).

[6] S. Ergün, B. Gültekin, and H. E. Ünalan, Percolation-based approach to resistive bridge modeling in metal nanowire networks on cracked substrates, Advanced Electronic Materials **11**, 2400840 (2025).

[7] J. W. Hutchinson and Z. Suo, Mixed mode cracking in layered materials, Advances in Applied Mechanics **29**, 63 (1992).

[8] M. D. Thouless, Crack spacing in brittle films on elastic substrates, Journal of the American Ceramic Society **73**, 2144 (1990).

[9] S. Bohn, L. Pauchard, and Y. Couder, Hierarchical crack pattern as formed by successive domain divisions. I. Temporal and geometrical hierarchy, Physical Review E **71**, 046214 (2005).

[10] J. Li and S.-L. Zhang, Finite-size scaling in stick percolation, Physical Review E **80**, 040104(R) (2009).

[11] I. Balberg and N. Binenbaum, Computer study of the percolation threshold in a two-dimensional anisotropic system of conducting sticks, Physical Review B **28**, 3799 (1983).

[12] Y. Y. Tarasevich and A. V. Eserkepov, Percolation thresholds for discorectangles: Numerical estimation for a range of aspect ratios, Physical Review E **97**, 042128 (2018).

[13] T. Hornig, I. M. Sokolov, and A. Blumen, Patterns and scaling in surface fragmentation processes, Physical Review E **54**, 4293 (1996).

[14] S. M. Bergin, Y.-H. Chen, A. R. Rathmell, P. Charbonneau, Z.-Y. Li, and B. J. Wiley, The effect of nanowire length and diameter on the properties of transparent, conducting nanowire films, Nanoscale **4**, 1996 (2012).

[15] A. P. Chatterjee and Y. Y. Tarasevich, Effect of polydispersity on the percolation threshold of sticks, Physical Review E **110**, 024137 (2024).

[16] F. Coupette and T. Schilling, Universality of the percolation threshold for polydisperse systems, Physical Review E **111**, 014107 (2025).

[17] K. Sampson, J. Forsman, and C. E. Woodward, Percolation of clustered, oriented, and polydisperse sticks, Physical Review E **107**, 044109 (2023).

[18] Y. Y. Tarasevich, A. V. Eserkepov, and S. Classes, Template-assisted percolation: experimental validation and conductivity of random nanowire networks on crack templates, Physical Review E **108**, 034134 (2023).